SAFAR

Phase 1

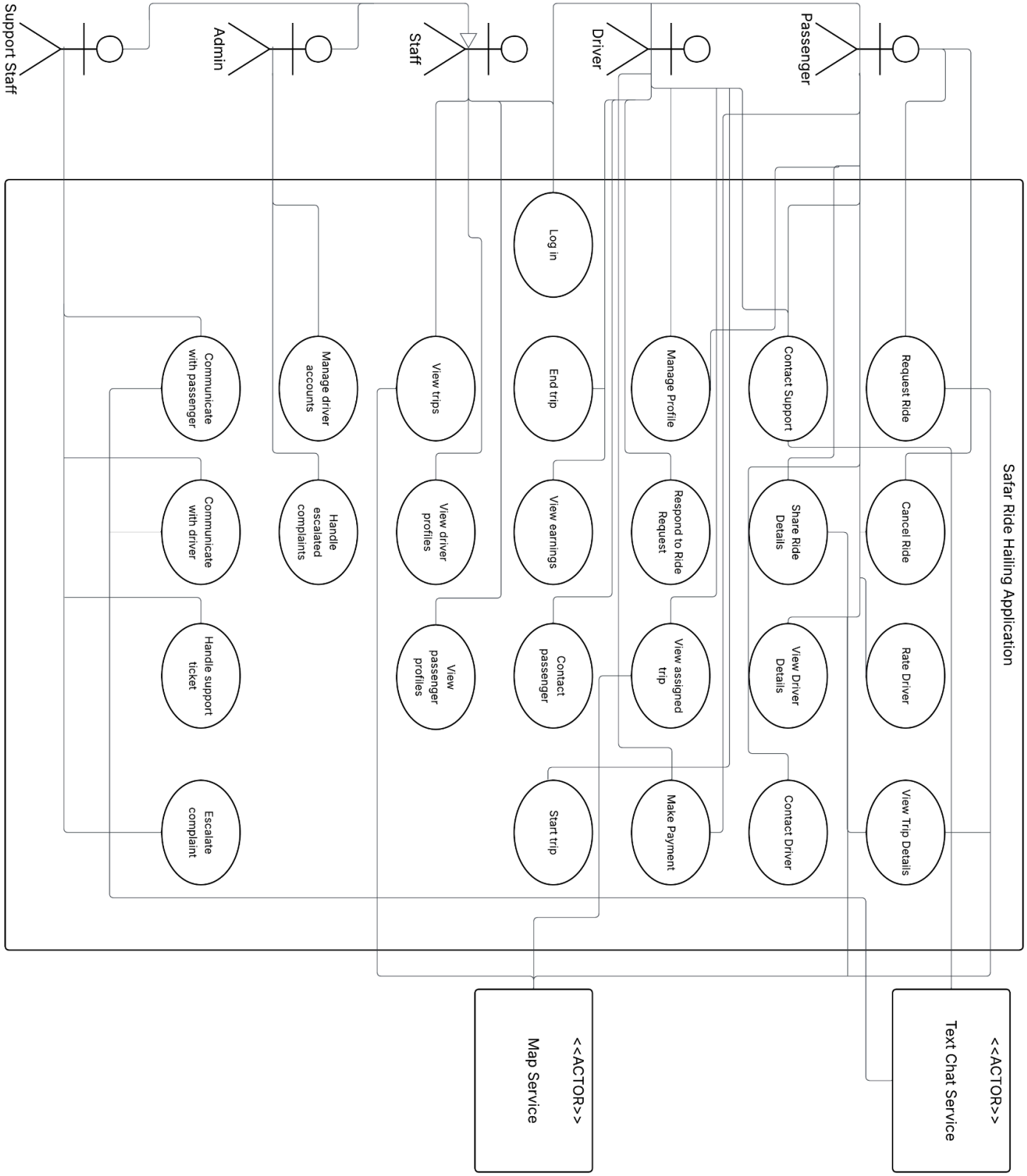
Team 1

Member Name	Member Roll #	Primary Responsibility
Syed Haider Ali	24L-0882	Use Case Diagram, Analysis Class Diagram, Use Case Descriptions 1, 17 - 26
Adnan Sami	24L-0803	Use Case Descriptions 10 - 16
Muhammad Luqman	24L-0736	Use Case Descriptions 2 - 9

TABLE OF CONTENTS

Use Case Diagram	**2**
Use Case Description 1	**3**
Use Case Description 2	**4**
Use Case Description 3	**5**
Use Case Description 4	**7**
Use Case Description 5	**8**
Use Case Description 6	**9**
Use Case Description 7	**10**
Use Case Description 8	**11**
Use Case Description 9	**12**
Use Case Description 10	**13**
Use Case Description 11	**14**
Use Case Description 12	**15**
Use Case Description 13	**16**
Use Case Description 14	**18**
Use Case Description 15	**19**
Use Case Description 16	**20**
Use Case Description 17	**21**
Use Case Description 18	**22**
Use Case Description 19	**23**
Use Case Description 20	**24**
Use Case Description 21	**25**
Use Case Description 22	**27**
Use Case Description 23	**28**
Use Case Description 24	**28**
Use Case Description 25	**30**
Use Case Description 26	**31**
Analysis Class Diagram	**33**

Use Case Diagram



Use Case Description 1

Identifier	UC001
Name	Log in
Summary	This use case describes the login procedure for users, where the user enters their credentials and is logged in by the system after verification
Priority	High
Actors	Passenger, Driver, Staff
Pre-condition(s)	None
Post-condition(s)	The user is logged in to their account
Dependencies	None

Typical Course of Action

S #	Actor Action	S #	System Response
1	User selects ‘Log in’	1.1	System displays login screen
2	User enters username and password and clicks confirm	2.1	The system authenticates the entered password
		2.2	If username is valid and password is correct, the user is logged in successfully

Alternate Course of Action

S #	Actor Action	S #	System Response
A1	Passenger selects ‘Sign up’	A1.1	System displays sign up screen
A2	Passenger enters a username and	A2	System creates a new user profile and logs the

	a password	.1	user in
B1	Actor tries to log in with a username that does not exist	B1 .1	System displays an error
C1	Actor tries to log in with incorrect password	C1 .1	System displays an error
Exceptions	None		

Use Case Description 2

Identifier	UC002
Name	Request Ride
Summary	This use case describes the procedure for a passenger to request a ride, which includes entering pickup and dropoff locations, selecting a preferred vehicle type, and confirming the ride request
Priority	High
Actors	Passenger, Driver
Pre-condition(s)	1 - Passenger account exists and the Passenger is logged in 2 - Passenger's device has an active internet connection 3 - Location services (GPS) are enabled on the Passenger's device
Post-condition(s)	The ride request is successfully submitted and broadcasted to nearby available Drivers
Dependencies	UC001 (Log in)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger opens the application and enters the desired drop-off location	1.1	System detects Passenger's current location via GPS and sets it as the default pickup location
		1.2	System calculates the route and displays available vehicle types along with estimated

			fares and arrival times for each
2	Passenger selects a preferred vehicle type and taps the confirm button	**2.1**	System validates the request
		2.2	System displays a looking for drivers screen
		2.3	System broadcasts the ride request to nearby available Drivers
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger enters drop-off location but decides to manually change the pickup location on the map	**1.1**	System updates the map to reflect the new pickup location
		1.2	System recalculates the route, estimated fares, and arrival times for all vehicle types based on the new locations
2	Passenger proceeds to select vehicle type and confirms	**2.1**	System continues with step 2.1 of the primary flow
Exceptions	**No Drivers Available:** System stops broadcasting, displays a "No drivers available" message, and prompts the Passenger to try again later if no Driver accepts the request within a predefined time limit **Location Not Found:** System displays an error message prompting the Passenger to re-enter or manually adjust the pin on the map if the entered pickup or dropoff location cannot be pinpointed		

Use Case Description 3

Identifier	UC003
Name	Cancel ride
Summary	This use case describes the process for a passenger to cancel a ride request, either while searching for a driver or after a driver has already accepted the trip

Priority	Medium
Actors	Passenger, Driver
Pre-condition(s)	1 - Passenger is logged in 2 - Passenger has an active ride request or an assigned driver 3 - The trip has not yet started
Post-condition(s)	The ride is canceled, and the Driver is notified if assigned
Dependencies	UC001 (Log in), UC002 (Request ride)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Passenger taps the cancel button on the looking for drivers screen	1	System displays a confirmation prompt
2	Passenger taps yes to confirm	2.1	System stops broadcasting the ride request to drivers
		2.2	System updates the ride status to canceled and returns the Passenger to the home screen

Alternate Course of Action

S #	Actor Action	S #	System Response
1	Passenger taps the cancel ride button on the active trip screen	1.1	System displays a confirmation prompt
2	Passenger taps yes to confirm	2.1	System cancels the active trip and immediately notifies the assigned Driver
		2.3	System returns the Passenger to the home screen

Exceptions	**Ride Already Started:** System disables the cancel ride button and displays a message advising the Passenger to ask the Driver to end the trip early or contact Support staff if the Driver has already swiped to start the trip

Use Case Description 4

Identifier	UC004
Name	Rate driver
Summary	This use case describes the process for a passenger to submit a rating for a driver after a ride is completed
Priority	Low
Actors	Passenger
Pre-condition(s)	1 - Passenger is logged in 2 - Driver has successfully ended the trip
Post-condition(s)	The rating and feedback are saved in the system and linked to the Driver's profile
Dependencies	UC001 (Log in)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Passenger views the rating prompt displayed after the trip ends	1.1	System displays the driver's name, photo, and a 1 to 5 star rating scale
2	Passenger selects a star rating	2.1	System highlights the selected stars and enables the submit button
3	Passenger types optional feedback and taps the submit button	3.1	System saves the rating to the database
		3.2	System displays a brief thank you confirmation
		3.3	System redirects the Passenger to the home screen

Alternate Course of Action

S #	Actor Action	S #	System Response

1	Passenger decides not to leave a rating and taps the skip button	1.1	System dismisses the rating prompt
		1.2	System redirects the Passenger to the home screen
Exceptions			

Use Case Description 5

Identifier	UC005
Name	Contact support
Summary	This use case describes the process for a passenger to reach out to support staff for assistance regarding a ride, account issues, or general inquiries
Priority	Medium
Actors	Passenger, Support staff
Pre-condition(s)	1 - Passenger is logged in 2 - Passenger's device has an active internet connection
Post-condition(s)	A support ticket or chat session is successfully created and routed to Support staff
Dependencies	UC001 (Log in)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger navigates to the help section in the application	1	System displays a list of common issues and a contact support button
2	Passenger taps the contact support button and selects the option to call support staff	2	System initiates a call with a support staff member
Alternate Course of Action			

S #	Actor Action	S #	System Response
1	Passenger chooses the live chat option instead of audio call	1	System opens a chat window and attempts to connect with Support staff
2	Passenger types a message and sends it	2	System connects the Passenger with Support staff and displays the real-time conversation
Exceptions	None		

Use Case Description 6

Identifier	UC006
Name	View trip details
Summary	This use case describes the process for a passenger to view information about an assigned trip including the driver's information, estimated time of arrival (ETA), and estimated fare
Priority	High
Actors	Passenger
Pre-condition(s)	1 - Passenger is logged in 2 - Passenger has an active ride that has been accepted by a Driver
Post-condition(s)	Passenger successfully views the real-time details and status of their active trip
Dependencies	UC001 (Log in), UC002 (Request ride)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Passenger views the active trip screen on the application	1.1	System displays the assigned Driver's name, photo, vehicle model, and license plate
		1.2	System calculates and displays the estimated time of arrival (ETA) to the pickup or dropoff location

		1.3	System calculates and displays the estimated fare for the trip
2	Passenger taps the screen to refresh the map view	**2.1**	System updates the Driver's real-time location on the map and adjusts the ETA accordingly
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger taps the estimated fare amount to see more information	**1**	System displays a detailed breakdown of the estimated fare including base rate and distance charges
2	Passenger dismisses the fare breakdown	**2**	System returns the Passenger to the main active trip screen
Exceptions	**Location Tracking Failure:** System displays a message indicating the Driver's real-time location is temporarily unavailable and shows the last known ETA if the Driver's GPS signal is lost		

Use Case Description 7

Identifier	UC007
Name	Share ride details
Summary	This use case describes the process for a passenger to share their live trip status, driver information, and estimated time of arrival with a contact
Priority	Low
Actors	Passenger
Pre-condition(s)	1 - Passenger is logged in 2 - Passenger has an active ongoing trip
Post-condition(s)	A tracking link containing the live trip details is successfully generated and sent to the selected contact
Dependencies	UC001 (Log in), UC002 (Request ride)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger taps the share ride button on the active trip screen	1	System generates a unique tracking link for the current trip
2	Passenger copies the message and returns to trip screen	2	System resumes displaying the active trip screen
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	None	1	None
Exceptions	**No Active Trip:** System disables or hides the share ride button if the trip has already ended or been canceled		

Use Case Description 8

Identifier	UC008
Name	View driver details
Summary	This use case describes the process for a passenger to view the profile and detailed information of their assigned driver
Priority	Low
Actors	Passenger
Pre-condition(s)	1 - Passenger is logged in 2 - Passenger has an assigned driver for an active trip
Post-condition(s)	Passenger successfully views the driver's profile information including ratings and vehicle details
Dependencies	UC001 (Log in), UC002 (Request ride)
Typical Course of Action	

S #	Actor Action	S #	System Response
1	Passenger taps on the driver's photo or name on the active trip screen	1	System displays the driver's profile including name, photo, overall rating, number of trips, and full vehicle details
2	Passenger reviews the information and taps the close or back button	2	System closes the driver profile and returns the Passenger to the active trip screen
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger taps on the driver's rating to view more information	1	System displays a list of recent anonymous feedback or badges earned by the driver
2	Passenger scrolls through the feedback and taps the back button	2	System returns the Passenger to the main driver profile screen
Exceptions			

Use Case Description 9

Identifier	UC009
Name	Contact driver
Summary	This use case describes the process for a passenger to communicate with their assigned driver via in-app chat or phone call regarding pickup details or delays
Priority	High
Actors	Passenger, Driver
Pre-condition(s)	1 - Passenger is logged in 2 - Passenger has an assigned Driver for an active trip
Post-condition(s)	Passenger successfully sends a message or initiates a phone call to the assigned Driver
Dependencies	UC001 (Log in), UC002 (Request ride)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger taps the chat icon on the active trip screen	1	System opens the in-app messaging interface
2	Passenger types a message and taps the send button	2.1	System transmits the message to the assigned Driver and displays it in the chat window
		2.2	System updates the message status to delivered or read when applicable
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger taps the phone icon on the active trip screen instead of chat	1	System prompts the Passenger to confirm they want to call the Driver
2	Passenger taps call to confirm	2	System initiates the call between the Passenger and Driver
Exceptions	**Ride Canceled or Ended:** System disables the chat and call icons and displays a message that the Driver can no longer be contacted if the trip is completed or canceled		

Use Case Description 10

Identifier	UC010
Name	Manage profile
Summary	This use case describes the procedure for a driver to view and update their profile information, including personal details, vehicle information, and profile picture.
Priority	Medium
Actors	Driver, Passenger

Pre-condition(s)	1. The actor's account exists and the actor is logged in 2. The actor's device has an active internet connection
Post-condition(s)	The actor's profile is successfully updated and saved
Dependencies	UC001(Log In)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Actor navigates to the profile section	**1.1**	System loads and displays the actor's current profile information
2	Actor edits the desired fields and taps 'Save'	**2.1**	System validates the updated information
		2.2	System saves the changes and displays a success confirmation message

Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Actor enters invalid information	**1.1**	Actor displays an error message

Exceptions	None

Use Case Description 11

Identifier	UC011
Name	Respond to Ride Request
Summary	This use case describes the procedure for a driver to view an incoming ride request and either accept or reject it.
Priority	High
Actors	Driver, Passenger
Pre-condition(s)	1. The Driver account exists and the Driver is logged in. 2. The Driver's device has an active internet connection.

	3. The Driver is online and available to accept rides
Post-condition(s)	The Driver has either accepted or rejected the ride request. If accepted, the Passenger is notified and trip begins
Dependencies	UC001 (Log In), UC002 (Request Ride)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Driver receives and views the incoming ride request	1.1	System displays ride request details including passenger pickup location, dropoff location and estimated fare
2	Driver taps "Accept"	2.1	System confirms the acceptance and notifies the Passenger
		2.2	System displays the Passenger's pickup location on the map for the Driver

Alternate Course of Action

S #	Actor Action	S #	System Response
1	Driver taps "Reject"	1.1	System cancels the request for this Driver and broadcasts it to other available Drivers

Exceptions	**Drivers unavailable:** System notifies the Passenger that no drivers are available

Use Case Description 12

Identifier	UC012
Name	View Assigned Trip
Summary	This use case describes the procedure for a driver to view the assigned trip details including the passenger's pickup and dropoff locations after accepting a ride request.
Priority	High

Actors	Driver
Pre-condition(s)	1. The Driver account exists and the Driver is logged in. 2. The Driver's device has an active internet connection. 3. The Driver has accepted a ride request.
Post-condition(s)	The Driver can clearly view the passenger's pickup and dropoff locations on the map.
Dependencies	UC001 (Log In), UC011 (Respond to Ride Request)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Driver navigates to the current trip screen	1.1	System displays the passenger's pickup and dropoff locations on the map
		1.2	System displays the recommended route and estimated arrival time to the pickup location
2	Driver reviews the trip details	2.1	System keeps the map updated in real time as the Driver moves

Alternate Course of Action

S #	Actor Action	S #	System Response
1	Driver loses internet connection while viewing trip details	1.1	System displays an error message

Exceptions	None.

Use Case Description 13

Identifier	UC013
Name	Start Trip
Summary	This use case describes the procedure for a driver to start the trip after arriving at the passenger's pickup location and confirming the passenger's presence.

Priority	High
Actors	Driver, Passenger
Pre-condition(s)	1. The Driver account exists and the Driver is logged in. 2. The Driver's device has an active internet connection. 3. The Driver has accepted a ride request and arrived at the pickup location
Post-condition(s)	The trip is successfully started and both the Driver and Passenger are notified.
Dependencies	UC001 (Log In), UC012 (View Assigned Trip)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Driver arrives at the passenger's pickup location and taps "Start Trip"	1.1	System verifies that pickup location is correct
		1.2	System starts the trip and begins tracking the route
		1.3	System notifies the Passenger that the trip has started
2	Driver begins driving towards the dropoff location	2.1	System tracks and updates the Driver's location in real time

Alternate Course of Action

S #	Actor Action	S #	System Response
1	Driver taps "Start Trip" but is at wrong pickup location	1.1	System shows an error message

Exceptions	None

Use Case Description 14

Identifier	UC014
Name	End Trip
Summary	This use case describes the procedure for a driver to end the trip after arriving at the passenger's dropoff.
Priority	High
Actors	Driver, Passenger
Pre-condition(s)	1. The Driver account exists and the Driver is logged in. 2. The Driver's device has an active internet connection. 3. A trip is currently in progress
Post-condition(s)	The trip is successfully completed ,fare is calculated and both the Driver and Passenger are notified.
Dependencies	UC001 (Log In), UC013 (Start Trip), UC026 (Make Payment)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Driver arrives at the passenger's dropoff location and taps "End Trip"	1.1	System verifies that dropoff location is correct
		1.2	System ends the trip and calculate the final fare
		1.3	System notifies the Passenger that the trip has ended
2	Driver views the trip summary	2.1	System displays the completed trip details including distance,fare,duration

Alternate Course of Action

S #	Actor Action	S #	System Response
1	Driver taps "Ends Trip" but is at wrong dropoff location	1.1	System shows an error message

2	Driver confirms ending the trip at current location		System ends the trip and calculates fare based on actual distance travelled
Exceptions	None		

Use Case Description 15

Identifier	UC015
Name	View Earnings
Summary	This use case describes the procedure for a driver to end the trip after arriving at the passenger's dropoff.
Priority	Medium
Actors	Driver
Pre-condition(s)	1. The Driver account exists and the Driver is logged in. 2. The Driver's device has an active internet connection. 3. The Driver has at least one completed trip.
Post-condition(s)	The Driver successfully views their earnings summary.
Dependencies	UC001 (Log In), UC014 (End Trip)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Driver navigates to the Earnings section	**1.1**	System retrieves and displays the Driver's total earnings
		1.2	System displays a breakdown of earnings per completed trip
2	Driver selects a specific trip to view details	**2.1**	System displays the detailed fare breakdown for selected trip
Alternate Course of Action			

S #	Actor Action	S #	System Response
1	Driver has no completed trip	1.1	System displays a message to Driver informing that no earnings are available yet
Exceptions	None		

Use Case Description 16

Identifier	UC016
Name	Contact Passenger
Summary	This use case describes the procedure for a driver to connect the Passenger via call or text during a request or trip.
Priority	Medium
Actors	Driver, Passenger
Pre-condition(s)	1. The Driver account exists and the Driver is logged in. 2. The Driver's device has an active internet connection. 3. The Driver has an accepted or ongoing trip with the Passenger.
Post-condition(s)	The Driver has successfully connected the Passenger via preferred communication method.
Dependencies	UC001 (Log In), UC011 (Respond to Ride Request)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Driver navigates to the trip screen and select ‘Contact Passenger’	1.1	System displays the available Contact options including call and text
2	Driver selects preferred contact method	2.1	System initiates the call or text interface with Passenger

3	Driver communicates with the Passenger	**3.1**	System logs the conversation for record
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger does not respond to call	**1.1**	System notifies Driver that Passenger is unavailable and suggests sending text instead
Exceptions	None		

Use Case Description 17

Identifier	UC017
Name	View trips
Summary	This use case describes how a staff member can view completed, cancelled and ongoing trips as well as their details
Priority	Medium
Actors	Staff
Pre-condition(s)	1. Actor is logged in 2. Actor's device has an active internet connection
Post-condition(s)	Actor views a list of trips, and views the details of selected trip(s)
Dependencies	UC001 (Log in)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Actor navigates to the 'Trips' tab	**1.1**	System displays a list of all completed,

			ongoing and cancelled rides
2	Actor selects a ride to view	2.1	System displays all details of that specific trip, such as pickup and dropoff locations, ride duration, driver profile and passenger profile
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Actor searches for specific trip using the search bar	1.1	System returns a list of matching trips and typical course of action resumes
2	Actors filters trips based on specific attributes such as date or status	2.1	System returns a list of all trips that match those attributes and typical course of action resumes
Exceptions	None		

Use Case Description 18

Identifier	UC018
Name	View driver profiles
Summary	This use case describes the procedure for admins and support staff to view the profiles of registered drivers
Priority	Medium
Actors	Staff
Pre-condition(s)	1. Actor is logged in 2. Actor has an active internet connection
Post-condition(s)	Actor views a list of all registered drivers, and views the details of selected driver(s)
Dependencies	UC001 (Log in)
Typical Course of Action	
S # / **Actor Action**	**S #** / **System Response**

1	Actor clicks 'Drivers'	**1.1**	System provides a list of all registered drivers
2	Actor selects a specific driver	**2.1**	System displays details of driver's profile, such as driver name and vehicle type
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Actor searches for specific driver using the search bar	**1.1**	System returns a list of matching driver profiles and typical course of action resumes
2	Actors filters driver based on specific attributes such as vehicle type		System returns a list of all drivers that match those attributes and typical course of action resumes
Exceptions	None		

Use Case Description 19

Identifier	UC019
Name	View passenger profiles
Summary	This use case describes the procedure for admins and support staff to view profile details of passengers
Priority	Medium
Actors	Staff
Pre-condition(s)	1. Actor is logged in 2. Actor has an active internet connection
Post-condition(s)	Actor views a list of all passengers, and views the details of selected passenger(s)
Dependencies	UC001 (Log in)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Actor clicks 'Passengers'	**1.1**	System provides a list of all passenger profiles

2	Actor selects a specific profile	**2.1**	System displays details of passenger's profile
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Actor searches for passenger using search box	**1.1**	System returns a list of matching passenger profiles, and typical course of action resumes
2	Actor filters passengers based on specific attributes	**2.1**	System returns a list of all passengers that match those attributes/conditions and typical course of action resumes
Exceptions	None		

Use Case Description 20

Identifier	UC020
Name	Manage driver accounts
Summary	This use case describes how an admin can manage the profiles of registered drivers, which includes creating and deleting driver accounts
Priority	High
Actors	Primary: Admin Secondary: Driver
Pre-condition(s)	1. Actor is logged in 2. Actor has an active internet connection
Post-condition(s)	Driver account information is created/updated/deleted successfully
Dependencies	UC001 (Log In), UC018 (View Driver Profiles)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Admin selects 'Drivers'	**1.1**	System displays a list of driver profiles
A1	Admin selects 'New Driver'	**A1 .1**	System displays a new driver form

A2	Admin enters required information and submits form	**A2.1**	System validates information
		A2.2	If info is valid, system creates new driver profile and displays confirmation message
B1	Admin selects a profile	**B1.1**	System displays driver profile
B2	Admin selects 'Edit'	**B2.1**	System displays editable driver profile
B3	Admin makes necessary changes to profile and submits	**B3.1**	System validates new information
		B3.2	If all new information is valid, system updates driver information and displays confirmation message
C1	Admin selects 'Delete' on a profile from drivers list	**C1.1**	System asks for deletion confirmation
C2	Admin confirms deletion	**C2.1**	System deletes selected driver profile and prints confirmation message
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
A1	Admin selects 'Delete' on a profile but cancels deletion	**A1.1**	System aborts deletion
B1	Admin leaves some required field empty while creating/editing a profile	**B1.1**	System displays error message and a prompt to fill missing information
Exceptions	None		

Use Case Description 21

Identifier	UC021
Name	Handle Escalated Complaints

Summary	This use case describes how an admin can respond to complaints escalated by support staff, which includes reviewing profiles and trips, taking administrative action and resolving complaints
Priority	Medium
Actors	Admin
Pre-condition(s)	1. Admin is logged in 2. Admin has an active internet connection 3. Support staff has escalated complaint(s)
Post-condition(s)	The admin has viewed the complaint and marked it as resolved
Dependencies	UC001 (Log in), UC018 (View driver profiles), UC019 (View passenger profiles), UC020 (Manage driver accounts), UC025 (Escalate Complaint)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Actor selects ‘Complaints’	1.1	System displays a list of complaints
2	Actor selects a complaint to view	2.1	System displays details of complaint such as driver profile, passenger profile and complaint description
3	Actor marks complaint as resolved	3.1	System updates complaint status and displays confirmation

Alternate Course of Action

S #	Actor Action	S #	System Response
1	After viewing complaint, actor views the related driver profile	1.1	System follows typical course of action for UC018
2	After viewing complaint, actor views the related passenger profile	2.1	System follows typical course of action for UC019
3	After viewing complaint, actor edits or deletes the related driver’s profile	3.1	System follows typical course of action for UC020 (section B or C)

Exceptions	**Passenger or driver profile no longer exists:** System displays an error message

Use Case Description 22

Identifier	UC022
Name	Communicate with passenger
Summary	This use case describes how support staff can contact passengers, which may be done as a response to their complaints
Priority	Medium
Actors	Primary: Support staff Secondary: Passenger
Pre-condition(s)	1. Support staff is logged in 2. Support staff has an active internet connection 3. Passenger has submitted a complaint
Post-condition(s)	Support staff has successfully contacted passenger via phone call
Dependencies	UC001 (Log in), UC024 (Handle support ticket)

Typical Course of Action

S #	Actor Action	S#	System Response
1	Support staff navigates to a complaint (UC024)	1.1	System displays complaint details (after following typical course of action for UC024)
2	Support staff selects ‘Contact passenger’	2.1	System dials passenger’s phone number
3	Support staff calls passenger		

Alternate Course of Action

S #	Actor Action	S #	System Response
1	None	1	None

Exceptions	None

Use Case Description 23

Identifier	UC023
Name	Communicate with driver
Summary	This use case describes how support staff can contact drivers, which may be done as a response to their complaints
Priority	Medium
Actors	Primary: Support staff Secondary: Driver
Pre-condition(s)	1. Support staff is logged in 2. Support staff has an active internet connection 3. Driver has submitted a complaint
Post-condition(s)	Support staff has successfully contacted driver via phone call
Dependencies	UC001 (Log in), UC024 (Handle support ticket)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Support staff navigates to a complaint (UC024)	1.1	System displays complaint details (after following typical course of action for UC024)
2	Support staff selects ‘Contact Driver’	2.1	System dials driver’s phone number
3	Support staff calls driver	3.1	None

Alternate Course of Action

S #	Actor Action	S #	System Response
1	None	1	None

Exceptions	None

Use Case Description 24

Identifier	UC024
Name	Handle support ticket
Summary	This use case describes how support staff can create, edit, delete and resolve support tickets, which are made after a passenger or driver contacts support staff.
Priority	High
Actors	Support Staff
Pre-condition(s)	1. Actor is logged in 2. Actor has an active internet connection 3. A passenger or driver has contacted support staff to register a complaint
Post-condition(s)	A support ticket has been created, edited, deleted or marked as resolved successfully
Dependencies	UC001 (Log in), UC005 (Contact support)

Typical Course of Action

S #	Actor Action	S #	System Response
1	Actor selects ‘Complaints’	**1.1**	System displays a list of complaints
A1	Actor selects complaint to View	**A1.1**	System displays the complaint detail
B1	Actor selects ‘New Ticket’	**B1.1**	System displays a ticket form containing fields such as passenger, driver, trip details
B2	Actor fills form and submits	**B2.1**	System creates new complaint ticket and displays confirmation message
C1	Actor selects ‘Delete’ on target ticket	**C1.1**	System displays prompt to confirm deletion
C2	Actor confirms deletion	**C2.1**	System deletes ticket and displays confirmation message
D1	Actor selects ‘Mark as Resolved’ on target ticket	**D1.1**	System displays prompt requesting confirmation

D2	Actor confirms action	**D2.1**	System marks ticket as resolved and displays confirmation
E1	Actor selects 'Edit' on target ticket	**E1.1**	System displays the filled ticket form, prompting updation
E2	Actor makes necessary changes and submits	**E2.1**	System saves updated version of complaint ticket and displays confirmation message
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Actor leaves field(s) in ticket form missing (while creating or editing ticket)	**1.1**	System displays an error message and returns to ticket, prompting actor to fill the required fields
2	Actor searches for a ticket instead of selecting it from the list	**2.1**	System returns all tickets that match the search, and typical course of action resumes
3	Actor filters out tickets based on certain conditions before selecting one	**3.1**	System returns a list of all tickets that match the entered criteria, and typical course of action resumes
Exceptions	**Action cancelled:** System returns and the requested action is not performed		

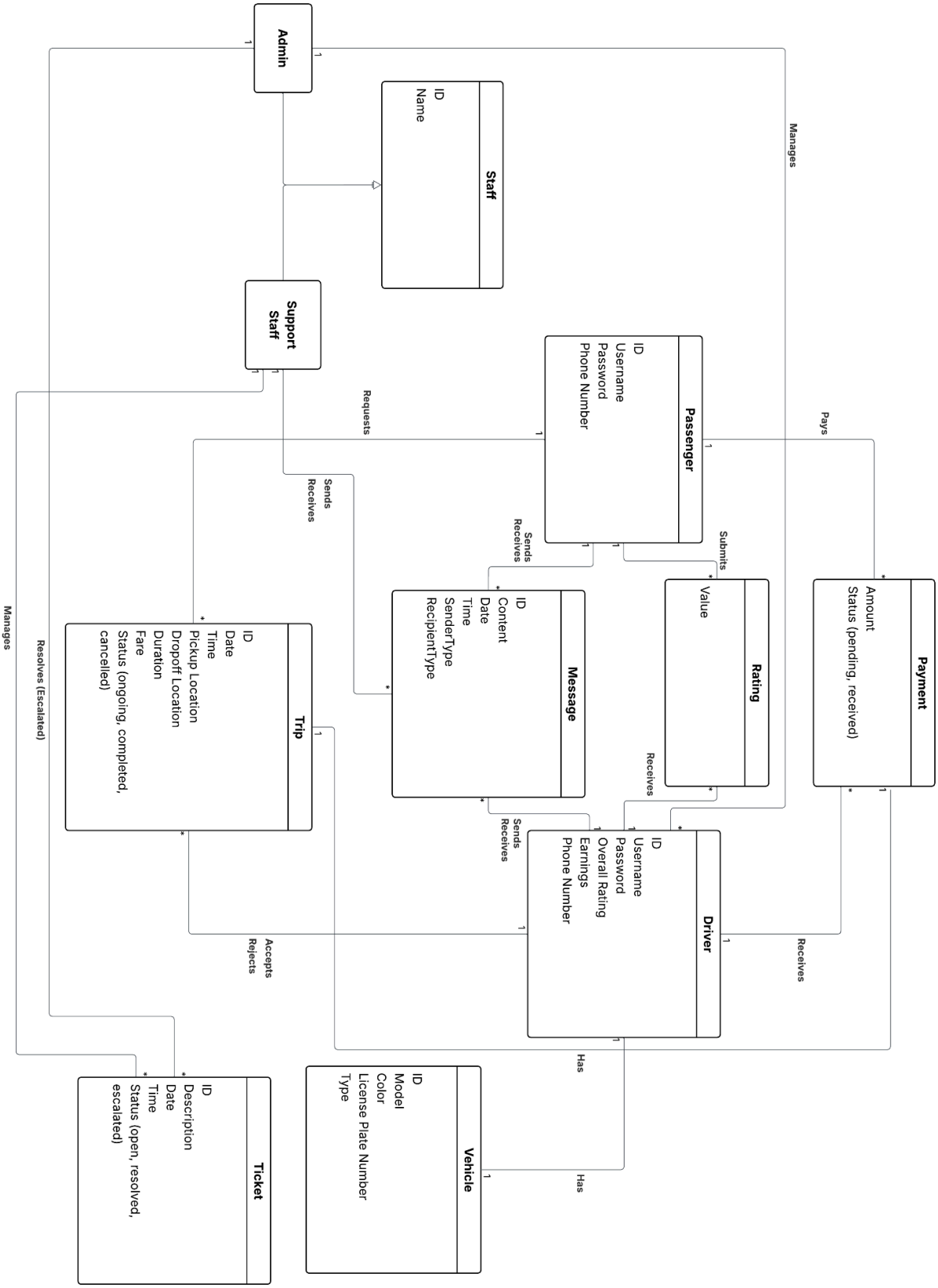
Use Case Description 25

Identifier	UC025
Name	Escalate complaint
Summary	This use case describes how support staff can escalate support tickets to admins
Priority	Medium
Actors	Support staff
Pre-condition(s)	1. Actor is logged in 2. Actor has an active internet connection 3. A support ticket exists

Post-condition(s)	The target support ticket has been forwarded to admins for review		
Dependencies	UC001 (Log in), UC024 (Handle support ticket)		
Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Actor selects 'Complaints'	**1.1**	System displays a list of tickets
2	Actor selects 'Escalate' on target complaint	**2.1**	System displays a prompt asking for confirmation
3	Actor confirms escalation	**3.1**	System forwards complaint to admins and displays confirmation message
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
	None		None
Exceptions	**Escalation not confirmed:** System does not escalate the ticket to admins		

Use Case Description 26

Identifier	UC026
Name	Make payment
Summary	This use case describes how a passenger pays the driver and the driver marks fare as paid
Priority	High
Actors	Driver, Passenger
Pre-condition(s)	1. Driver is logged in 2. Driver has an active internet connection
Post-condition(s)	The fare of the trip will be paid
Dependencies	UC001 (Log in)

Typical Course of Action			
S #	**Actor Action**	**S #**	**System Response**
1	Passenger pays fare to driver and driver selects ‘Payment Received’	**1.1**	System marks the payment as complete
Alternate Course of Action			
S #	**Actor Action**	**S #**	**System Response**
	None		None
Exceptions	None		

Analysis Class Diagram